Aligning RGB Channels (using SSD) of the given image to produce colored and properly aligned and clear image

Asked 3 years, 5 months ago Modified 2 years, 8 months ago Viewed 5k times

2

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The problem statement is -
To search over a window of possible displacements (You will search [-10,10] pixels), score each one using some image matching metric, and take the displacement with the best score. The simplest one is the Sum of Squared Differences (SSD) distance, which simply subtracts one image region from the other and evaluates the sum of the squared values in each pixel. You need to do SSD based calculations on image regions which are of double type.
As R,G,B channels are highly correlated to each other, SSD metric is very likely to work. Your task is to hold the G channel as the reference and move around R and B channels over it and keep track of the SSD value for the 51x51 image regions at the center of the channels. For each channel, you will find the (x,y) displacement vector that gives the minimum SSD value.

The program should divide the image into three equal parts.
The first image is Blue Channel, the second one is Green channel and the last one is Red channel. You have to name your variables as below

Blue channel Image - B
Green channel Image - G
Red channel Image - R

Reference Green Channel center region (51x51) -
ref_img_region (center of this region coincides with the center of image and indexes are always integers) Final aligned image - ColorImg_aligned

Given input image-

Required output image- 

I have done this so far and getting the almost same output as required but the compiler is showing incorrect value for ColorImg_aligned.

Previous Assessment: 5 of 6 Tests Passed (57%)

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Validating B	15%	(15%)
Validating R	14%	(14%)
Validating G	14%	(14%)
Validating 51x51 Green channel reference sub-image region	14%	(14%)
Usage of im2double	0%	(0%)
Validate Aligned Color Image	0%	(43%)

Variable ColorImg_aligned has an incorrect value.

The aligned color image looks like this

There are many places your code may have gone wrong. We suggest you re-check your code for errors

Total: 57%

Output

What's wrong with my code. I am new to Matlab. Sorry if the code is not efficient.

```
%Read the image
img = imread('course1image.jpg');
[r,c] = size(img);

B = img(1:r/3,:);
G = img((r/3)+1:(2*r/3),:);
R = img((2*r/3)+1:r,:);

ref_img_region = G;
[rg,cg] = size(ref_img_region);
ref_img_region = ref_img_region(ceil((rg-50)/2) :ceil((rg-50)/2) + 50,ceil((cg-50)/2)
:ceil((cg-50)/2) + 50);
%disp(size(ref_img_region));
ref_img_region = double(ref_img_region);

% Naive way
% ColorImg_aligned = cat(3,R,G,B);
% imshow(ColorImg_aligned);

% SSD way
nR = align(G,R);
```

```
nB = align(G,B);
ColorImg_aligned = cat(3,nR,G,nB);
imshow(ColorImg_aligned);


function aligned = align(green,red)
    [red_row,red_col] = size(red);
    [green_row,green_col] = size(green);

    % checking SSD for cropped part of the images for faster calculation
    cropped_red = red(ceil((red_row-50)/2) : ceil((red_row-50)/2) + 50,ceil((red_col-
50)/2) :ceil((red_col-50)/2) + 50);
    cropped_green = green(ceil((green_row-50)/2) : ceil((green_row-50)/2) +
50,ceil((green_col-50)/2) :ceil((green_col-50)/2) + 50);

    MiN = 9999999999;
    r_index = 0;
    r_dim = 1;
    for i = -10:10
        for j = 1:2
            ssd =     SSD(cropped_green,circshift(cropped_red,i,j));
            if ssd < MiN
                MiN = ssd;
                r_index = i;
                r_dim = j;
            end
        end
    end
    aligned = circshift(red,r_index,r_dim);
end

function ssd = SSD(a1,a2)
    x = double(a1)-double(a2);
    ssd = sum(x(:).^2);
end
```

matlab image-processing rgb Edit tags

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edited Jul 10, 2019 at 15:19

asked Jul 10, 2019 at 14:22

Ashish kumar
173 1 4 10

4 Answers

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1

Finally I was able to pass the assignment.

I just changed

Y = circshift(A,K,dim)

version of circshift to

Y = circshift(A,[i,j])

I'm still doing the same thing but the compiler accepted the answer this time. Anybody having thoughts about why this happened?

After modification the code looks like this.

```
%Read the image
img = imread('course1image.jpg');
[r,c] = size(img);

B = img(1:r/3,:);
G = img((r/3)+1:(2*r/3),:);
R = img((2*r/3)+1:r,:);

ref_img_region = G;
[rg,cg] = size(ref_img_region);
ref_img_region = ref_img_region(ceil((rg-50)/2) :ceil((rg-50)/2) + 50,ceil((cg-50)/2)
:ceil((cg-50)/2) + 50);
%disp(size(ref_img_region));
ref_img_region = double(ref_img_region);

% Naive way
% ColorImg_aligned = cat(3,R,G,B);
% imshow(ColorImg_aligned);

% SSD way
nR = align(G,R);
nB = align(G,B);
ColorImg_aligned = cat(3,nR,G,nB);
imshow(ColorImg_aligned);


function aligned = align(green,red)
    [red_row,red_col] = size(red);
    [green_row,green_col] = size(green);

    % checking SSD for cropped part of the images for faster calculation
    cropped_red = red(ceil((red_row-50)/2) : ceil((red_row-50)/2) + 50,ceil((red_col-
50)/2) :ceil((red_col-50)/2) + 50);
cropped_green = green(ceil((green_row-50)/2) : ceil((green_row-50)/2) +
50,ceil((green_col-50)/2) :ceil((green_col-50)/2) + 50);

    MiN = 9999999999;
    r_index = 0;
    r_dim = 1;
    % Modifications
    for i = -10:10
        for j = -10:10
            ssd =
SSD(cropped_green,circshift(cropped_red,[i,j])); %circshift(A,[i,j])
            if ssd < MiN
                MiN = ssd;
                r_index = i;
                r_dim = j;
            end
        end
    end
    aligned = circshift(red,[r_index,r_dim]);
end

function ssd = SSD(a1,a2)
    x = double(a1)-double(a2);
    ssd = sum(x(:).^2);
end
```

Output

Assessment: All Tests Passed (100%)

Submit

Validating B	15% (15%)
Validating R	14% (14%)
Validating G	14% (14%)
Validating 51x51 Green channel reference sub-image region	14% (14%)
Usage of im2double	0% (0%)
Validate Aligned Color Image	43% (43%)

Total: 100%

Output

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answered Jul 11, 2019 at 20:02

Ashish kumar
173 1 4 10

-1

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Please help! I can't pass this task, my code nor your code acctually doesn't work What am I missing in the solution?

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answered Apr 24, 2020 at 8:39

Kate
1

1 This does not provide an answer to the question. You can search for similar questions, or refer to the related and linked questions on the right-hand side of the page to find an answer. If you have a related but different question, ask a new question, and include a link to this one to help provide context. See: Ask questions, get answers, no distractions – Natty Apr 24, 2020 at 8:40

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0

This post is hidden. It was deleted 2 years ago by Bhargav Rao.

enter image description hereenter image description here

i am still getting this error, can anyone please help me?

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edited Apr 23, 2020 at 15:42

answered Apr 23, 2020 at 15:41

Muhammad Nabeel
1 1

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This post is hidden. It was deleted 2 years ago by Samuel Liew ♦.

I tried to run your code but have received this error on the align function any idea what could be the reason?

> RGB = cat(3, R, G, B); imshow(RGB) aR = align(R,B); aG = align(G,B); Error using align (line 107) Wrong number of input arguments for ALIGN

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answered Jan 15, 2020 at 6:10

arman kami
1

This does not provide an answer to the question. Once you have sufficient reputation you will be able to comment on any post; instead, provide answers that don't require clarification from the asker. - From Review – sud007 Jan 15, 2020 at 6:38

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